RESCUE JUNIOR - Bill of Materials (BOM)

Team name: Sleeping

Every Line/Maze team has to submit a bill of materials for their robot

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

Team name:	Sleeping

#	Component	Part name	Autor	Source
1	PCB with electronic components	Custom made	Name of team members/Jlcpcb	jlcpcb.com
2	Wheels	Nexus Robot Mecanum Wheel	Nexus Robot	robotshop.com
3	Sheet metall parts for frame	Custom made		Local workshop
4	Drive motors	Maxon ECX TORQUE 22 XL Ø22 mm	Maxon Group	Maxon Motors
5	Robots shell	Custom made	Name of team members	3D Printer
6	Robots costume	Custom made	Name of team members	Self made
7	Microcontroller	Arduino Mega 2560	Arduino	conrad.de
8	Connecting parts	Screws, nuts, washers, ...		Local tool store
9	Motors for arm	NEMA-17-01 NEMA-17-01 0.4 Nm 1.68 A	Conrad Electronic	conrad.de
10	Stepper motor controllers	Pololu Tic T825	Electronics	conrad.de
11	LEDs	Various colours and sizes	Farnell An Avnet Company	farnell.com
12	Battery	11.1 V 3700 mAh	Farnell An Avnet Company	farnell.com
13	Cables	Various cables		Local electronics store
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Name of the Local Currency	Total Cost Local Currency	Total Cost ($) U.S.A. Dollars
Brazilian Real	R$11,824.09	$2,094.00

Quantity	Unit cost (Local	Unit cost (USA Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	R$338.80	$60.00	R$338.80	$60.00
4	R$112.93	$20.00	R$451.72	$80.00
1	R$1,129.34	$200.00	R$1,129.34	$200.00
4	R$1,694.01	$300.00	R$6,776.04	$1,200.00
1	R$112.93	$20.00	R$112.93	$20.00
1	R$282.34	$50.00	R$282.34	$50.00
2	R$237.16	$42.00	R$474.32	$84.00
1	R$225.87	$40.00	R$225.87	$40.00
2	R$118.58	$21.00	R$237.16	$42.00
2	R$508.20	$90.00	R$1,016.40	$180.00
20	R$2.82	$0.50	R$56.40	$10.00
3	R$203.28	$36.00	R$609.84	$108.00
1	R$112.93	$20.00	R$112.93	$20.00

Team name:	Sleeping

#	Name	Software's Tool/Library	Description
1	VL53L0X-Arduino	Library	For controlling the VL53L0X distance sensor on Arduino
2	Open CV	Library	Image processing library
3	IDE Arduino	Software's Tool	Software
4	Visual Studio Code	Software's Tool	Software
5	Numpy	Library	Array and math library
6	Matplotlib	Library	Python graphics generation library
7	Tinckercad	Software's Tool	Online platform for simulating circuits
8	Git	Software's Tool	Version control for managing functions
9	Fusion 360	Software's Tool	3D CAD shell design
10	Tic Arduino	Library	Library for controlling a Nema 17 stepper motor with the Tic T825 driver
11	Modulemotorl298n	Library	module without having to worry about the pins manually in each sketch.
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Name of the Local Currency	Total Cost Local Currency	Total Cost ($) U.S.A. Dollars
Brazilian Real	R$0.00	$0.00

Source	Autor	Currency)	Dollars)
pololu/vl53l0x-arduino: Biblioteca P	Pololu	FREE	FREE
OpenCV - Biblioteca abierta de Co	Opencv.org	FREE	FREE
Software | Arduino	Arduino.cc	FREE	FREE
Visual Studio Code: edición de cód	Microsoft	FREE	FREE
NumPy	Numpy developers	FREE	FREE
Matplotlib — Visualización con Pyt	Matplot lib dev team	FREE	FREE
Tinkercad	Autodesk	FREE	FREE
Git - Descargas	Software Freedom Conservancy	FREE	FREE
Fusión de Autodesk | Software bas	Autodesk	FREE/EDUCATION	FREE/EDUCATION
https://github.com/pololu/tic-arduin	DavidEGrayson	FREE	FREE
	Name of team members	FREE	FREE

Team name:	Sleeping

#	Component	Part name	Autor	Source	Quantity
1	main controller	raspberry pi 5	raspberry pi	e.tb.cn	1
2	micro controller	arduino nano mini	arduino	e.tb.cn	1
3	motor driver	tb6612fng	Toshiba	e.tb.cn	2
4	google coral edge tpu	google coral edge tpu	Google	www.mo	1
5	ultrasonic sensor	HC-SR04		e.tb.cn	2
6	vl53l0x	vl53l0x		e.tb.cn	2
7	connecter	HT508V		e.tb.cn	12
8	battery	18650		e.tb.cn	3
9	servo	servo		e.tb.cn	4
10	imu sensor	bno055	DFRobot	e.tb.cn	1
11	wheel	Custom made			4
12	pcb	Custom made	LCPCB	lceda.cn	3
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Name of the Local Currency	Total Cost Local Currency	Total Cost ($) U.S.A. Dollars
Brazilian Real BRL	R$4,374.60	$541.98

(Local	Unit cost (US Dollars)	(Local	(USA Dollars)
R$3,099.00	$384.00	R$3,099.00	$384.00
R$17.00	$2.10	R$17.00	$2.10
R$7.00	$0.86	R$14.00	$1.72
R$729.00	$90.30	R$729.00	$90.30
R$5.00	$0.61	R$10.00	$1.22
R$12.00	$1.49	R$24.00	$2.97
R$0.30	$0.04	R$3.60	$0.45
R$15.00	$1.86	R$45.00	$5.58
R$16.00	$1.98	R$64.00	$7.93
R$139.00	$17.22	R$139.00	$17.22
R$20.00	$2.48	R$80.00	$9.91
R$50.00	$6.20	R$150.00	$18.59
	$0.00	R$0.00	$0.00
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Team name:	Sleeping

#	Name	Software's Tool/Library	Description	Source
1	opencv	Library	computer vision	
2	numpy	Library	math	
3	vl53l0x	Library	getting distence	
4	bno055	Library		
5	yolo	Library	object detection	
6	vs code	software	programming	
7	shapr3d	software	CAD design	
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Name of the Local Currency	Total Cost Local Currency	Total Cost ($) U.S.A. Dollars
Brazilian Real	R$0.00	$0.00

Autor	**Currency)**	**Dollars)**
	Free	Free
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	Free	Free
	Free	Free
	Free	Free
Microsoft	Free	Free
	Free	Free

